

Cluster Analysis Interpretation

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All data preparation, PCA, and cluster analysis were carried out in Python. The raw dataset was first cleaned by handling missing, irregular, and outlying values to improve the quality of the data. Categorical variables were then regrouped and encoded into formats suitable for analysis in Python. Additional variables were also derived to better represent customer characteristics and purchasing behavior, while redundant variables were removed. Before PCA, selected variables were transformed to address skewness and then standardized so that all features were measured on a comparable scale. After these preprocessing steps in Python, the refined dataset was used for PCA to reduce dimensionality and for cluster analysis to identify customer segments.

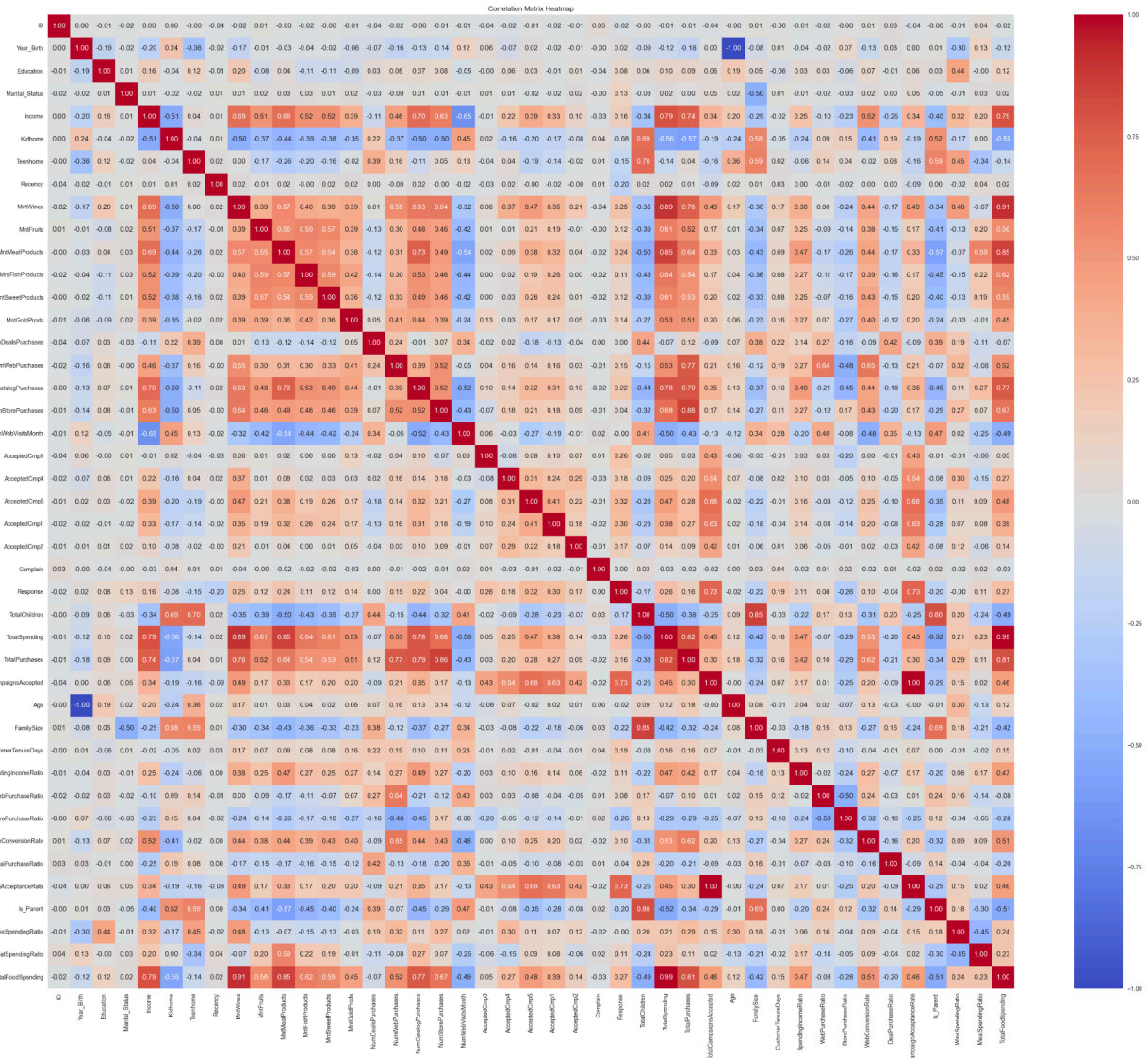
The feature-engineered variables created during preprocessing were:

1. TotalChildren
2. TotalSpending
3. TotalPurchases
4. TotalCampaignsAccepted
5. Age
6. FamilySize
7. CustomerTenureDays
8. SpendingIncomeRatio
9. WebPurchaseRatio
10. StorePurchaseRatio
11. WebConversionRate
12. DealPurchaseRatio
13. CampaignAcceptanceRate
14. Is_Parent
15. WineSpendingRatio
16. MeatSpendingRatio
17. TotalFoodSpending

The analysis ended up yielding **2 components** and **3 clusters**.

Principal Component Analysis

PCA was necessary because the dataset contained many features, and the correlation matrix showed that several of them were strongly related to one another.



This indicated redundancy in the information carried by the original variables made the dataset more complex to analyze directly. PCA helped reduce the number of features by transforming the correlated variables into a smaller set of principal components while still preserving most of the variation in the data. This made the dataset easier to interpret, simpler to visualize, and more manageable for further analysis. It also provided a more reliable input for cluster analysis by reducing multicollinearity and focusing on the most important patterns in the data.

The variables included in the final PCA model were selected through both logical assessment and computational testing in Python. First, variables were reviewed based on their sample proportions, histogram distributions, and established relevance to customer segmentation in similar datasets, ensuring that the retained features were both interpretable and meaningful. In addition, a custom Python algorithm was developed to brute-force test all possible PCA model combinations within a reasonable range.

Considerable iterative tuning and experimentation were also performed throughout the process, with variables being repeatedly added, removed, transformed, and reassessed to refine the candidate feature set and improve overall model quality.

A full brute-force search across every possible variable combination was not feasible because of computational limitations, so the search space was narrowed to practical candidate sets. This combined approach allowed the final PCA model to be chosen not only on theoretical and business logic grounds, but also on systematic evidence from model exploration.

The Python brute-force search was limited to the following candidate variables:

1. Income
2. NumWebVisitsMonth
3. TotalSpending
4. TotalPurchases
5. Age
6. FamilySize
7. SpendingIncomeRatio
8. StorePurchaseRatio
9. WebConversionRate
10. DealPurchaseRatio
11. WineSpendingRatio
12. MeatSpendingRatio
13. CampaignAcceptanceRate

Other variables that were not directly included in the PCA and clustering procedures were retained for the descriptive interpretation of the customer segments. Descriptive and inferential statistics across these additional variables were later examined to provide a more detailed characterization and business interpretation of each identified customer group.

```
Components: 2
Rotation: oblimin
Overall KMO: 0.841
Minimum item KMO: 0.788
Bartlett's test: chi-square = 15844.328, p < .001
Explained variance: 0.814
```

The brute-force procedure selected a two-component solution with oblimin rotation, and the resulting model appears statistically strong and well-defined.

Sampling adequacy was high, as indicated by an overall Kaiser–Meyer–Olkin (**KMO**) **measure of 0.841**, suggesting that the variables shared sufficient common variance for component extraction to be appropriate. The minimum item-level KMO of 0.788 further indicates that this adequacy was consistent across the retained variables, with no item demonstrating problematic fit for inclusion in the model.

This was further supported by Bartlett's test of sphericity, $\chi^2 = 15844.328$, **p < .001**, indicating that the correlation matrix significantly differed from an identity matrix and that the variables were sufficiently intercorrelated to justify PCA.

The retained two-component solution explained **81.4% of the total variance**, which is substantial for a parsimonious model of this size.

	item_kmo
NumWebVisitsMonth	0.8092
TotalSpending	0.8096
TotalPurchases	0.9014
FamilySize	0.7882
SpendingIncomeRatio	0.8079
WebConversionRate	0.8733
DealPurchaseRatio	0.8595

The item-level KMO values indicate that all retained variables demonstrate acceptable to excellent sampling adequacy for PCA, with values ranging from 0.788 to 0.901. Notably, **TotalPurchases exhibited the strongest adequacy (KMO = 0.901)**, suggesting that it shares a particularly high proportion of common variance with the other variables in the model.

Meanwhile, FamilySize showed the lowest value (KMO = 0.788), though it still exceeds the commonly accepted threshold of 0.70 and therefore remains suitable for inclusion.

The consistently high item-level KMO values suggest that the retained variables are **sufficiently interrelated and contribute meaningfully to the shared component structure**, with no variable demonstrating problematic sampling adequacy.

	communality
NumWebVisitsMonth	0.4844
TotalSpending	0.9560
TotalPurchases	0.9678
FamilySize	0.8322
SpendingIncomeRatio	0.8968
WebConversionRate	0.7567
DealPurchaseRatio	0.5949

The communalities indicate that the retained variables were **generally well represented** by the two-component solution, with values ranging from 0.484 to 0.968.

Most variables demonstrated strong representation, particularly TotalPurchases (0.968) and TotalSpending (0.956), suggesting that a substantial proportion of their variance was captured by the extracted components. Overall, the communalities support the adequacy and interpretability of the final PCA model.

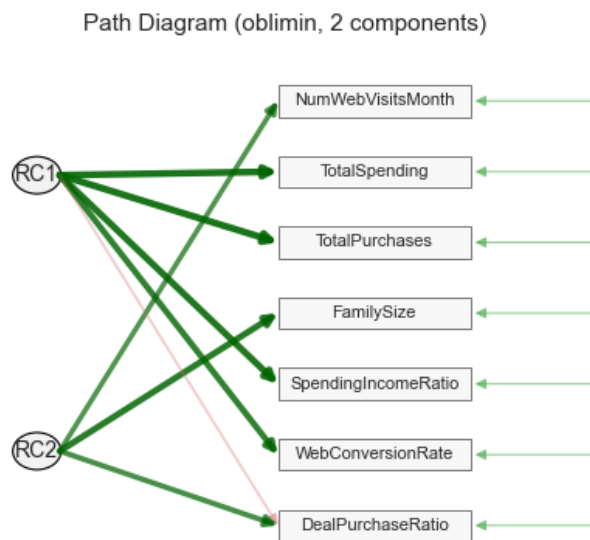
The component loadings indicate a clear and strong simple structure, with all retained variables loading substantially onto their respective components and **no problematic cross-loadings observed**.

PC1 was primarily defined by TotalPurchases (0.983), TotalSpending (0.978), SpendingIncomeRatio (0.947), and WebConversionRate (0.860).

Meanwhile, PC2 was strongly associated with FamilySize (0.902), DealPurchaseRatio (0.732), and NumWebVisitsMonth (0.684).

	PC1	PC2
NumWebVisitsMonth		0.6838
TotalSpending	0.9778	
TotalPurchases	0.9827	
FamilySize		0.9016
SpendingIncomeRatio	0.947	
WebConversionRate	0.8604	
DealPurchaseRatio		0.7322

The **high loadings** and absence of cross-loadings support the interpretability and discriminant clarity of the final two-component solution.



The path diagram visually reinforces the presence of a **clean and interpretable** two-component structure.

RC1 demonstrates strong positive relationships with TotalSpending, TotalPurchases, SpendingIncomeRatio, and WebConversionRate, suggesting that this component primarily reflects **customer purchasing power**.

In contrast, RC2 is more strongly associated with FamilySize, DealPurchaseRatio, and NumWebVisitsMonth, indicating a component related to **household-driven and deal-oriented purchasing behavior**.

Cluster Analysis

Following PCA, the retained components were used as the input variables for customer segmentation. Using the transformed components instead of the raw variables allowed the clustering process to operate on a cleaner and more compact representation of the data, reducing redundancy while preserving the majority of the shared variance structure. To identify the optimal clustering configuration, a Python-based brute force search algorithm was implemented to evaluate multiple clustering models and parameter combinations. Based on the evaluation metrics, the best-performing solution was a 3-cluster model using the K-means clustering algorithm, which demonstrated strong separation and interpretability across customer groups.

Clusters	N	R ²	AIC	BIC	Silhouette
3	2208	0.740	12435	12486	0.453

This clustering solution suggests that the retained 3-cluster model demonstrates **strong overall performance and separation quality**. The model explained 74.0% of the variance ($R^2 = 0.740$), indicating that the clusters captured a substantial proportion of the structure within the transformed feature space.

The **silhouette coefficient of 0.453** further suggests moderate-to-strong cluster separation, indicating that observations were generally well matched to their assigned clusters relative to neighboring clusters.

Additionally, the **relatively lower AIC (12435) and BIC (12486) values** support the adequacy and efficiency of the selected clustering structure. These results indicate that the final 3-cluster solution achieved a strong balance between interpretability, compactness, and cluster separation.

Davies-Bouldin	Balance	Algorithm	Score
0.810037	0.710112	kmeans	1.1

The final clustering model achieved a **Davies–Bouldin Index of 0.810**, indicating relatively good cluster separation and compactness, as lower values suggest better-defined clusters.

The **balance score of 0.710** further indicates that the cluster sizes were reasonably distributed, reducing the likelihood of highly uneven or disproportionately small clusters.

The cluster proportions further support the balance metric, with Cluster 2 containing 40.31% of observations, Cluster 3 containing 31.07%, and Cluster 1 containing 28.6%. This **relatively even distribution** suggests that the final K-means solution avoided severe cluster imbalance, supporting the stability and practical interpretability of the segmentation structure.

Cluster	Proportion
1	0.286232
2	0.403080
3	0.310688

	1	2	3
Cluster			
Size	632	890	686
Explained proportion within-cluster heterogeneity	0.272	0.402	0.325
Within sum of squares	1172.0	1730.1	1400.0
Silhouette score	0.465	0.520	0.355

The cluster-level metrics suggest that all three clusters demonstrated acceptable cohesion and separation.

Specifically, **Cluster 2 exhibited the strongest overall clustering quality**. It achieved the highest silhouette score (0.520) and accounted for the largest proportion of within-cluster heterogeneity (40.2%), indicating a well-defined and information-rich segment. Cluster 1 also demonstrated strong separation (silhouette = 0.465), while Cluster 3 showed comparatively weaker but still acceptable clustering quality (silhouette = 0.355).

The silhouette scores indicate that the clusters were **reasonably distinct and internally cohesive**.

	Value
Maximum diameter	8.271
Minimum separation	0.052
Pearson's γ	0.660
Dunn index	0.006
Entropy	1.087
Calinski- Harabasz index	3140

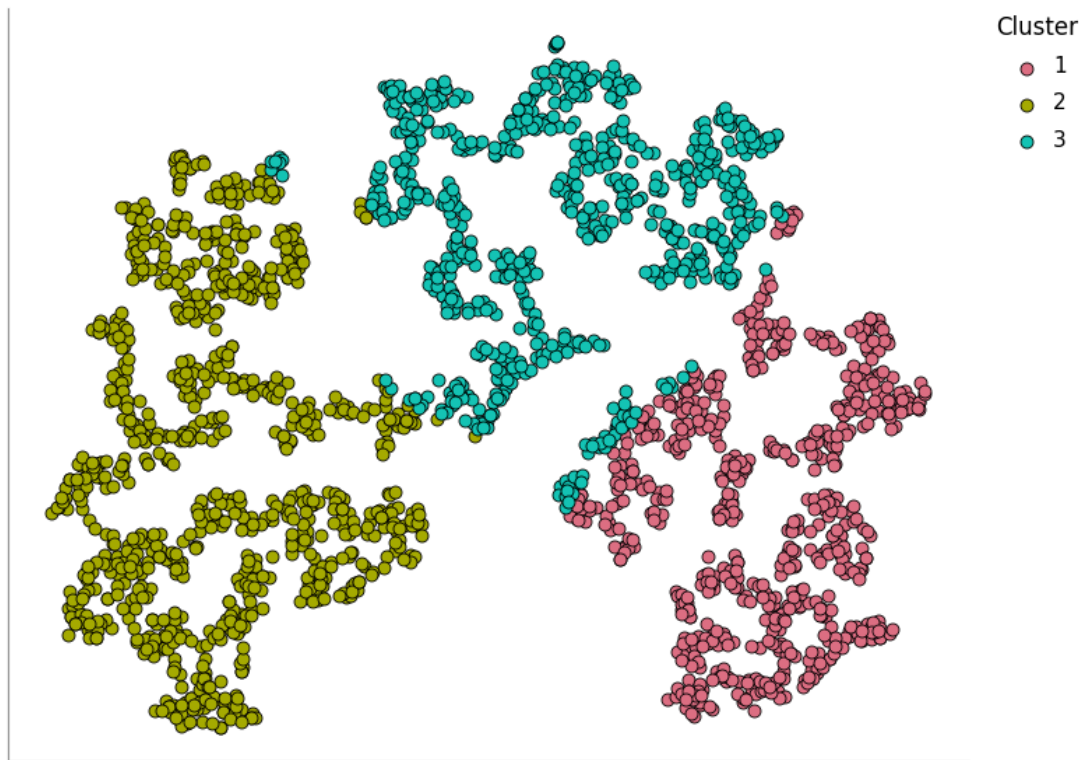
Using Euclidean distance as the distance metric, the clustering solution demonstrated mixed but **generally acceptable structural properties**.

The Calinski–Harabasz index (3140) and Pearson's γ (0.660) suggest relatively strong cluster separation and internal cohesion.

Meanwhile, the entropy value (1.087) indicates a **reasonably distributed clustering structure** across the three groups.

However, the low Dunn index (0.006) and minimum separation value (0.052) suggest that certain cluster boundaries remain relatively close to one another despite the overall adequacy of the solution. These metrics indicate that the final K-means model achieved a practically interpretable and statistically acceptable segmentation structure under Euclidean distance.

t-SNE Cluster Plot



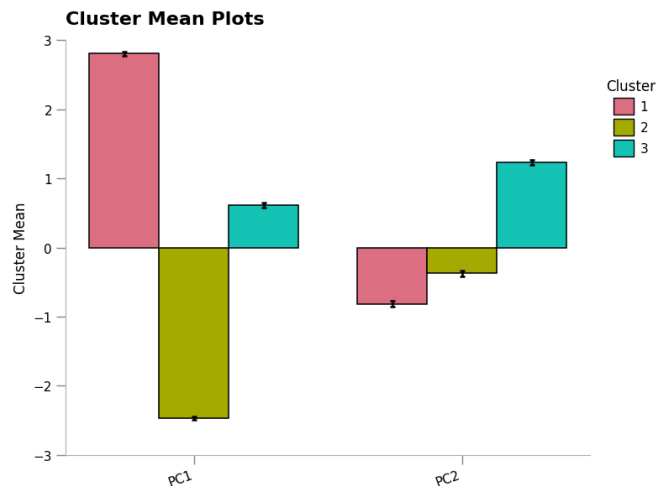
The t-SNE visualization further supports the clustering metrics by providing a visual representation of the separation among the three customer segments.

Distinct grouping patterns can be observed, with most observations clustering closely within their assigned groups and **relatively limited overlap between clusters**.

In particular, the visualization supports the moderate-to-strong silhouette scores observed in the clustering metrics, as the clusters appear reasonably compact and distinguishable within the reduced-dimensional space.

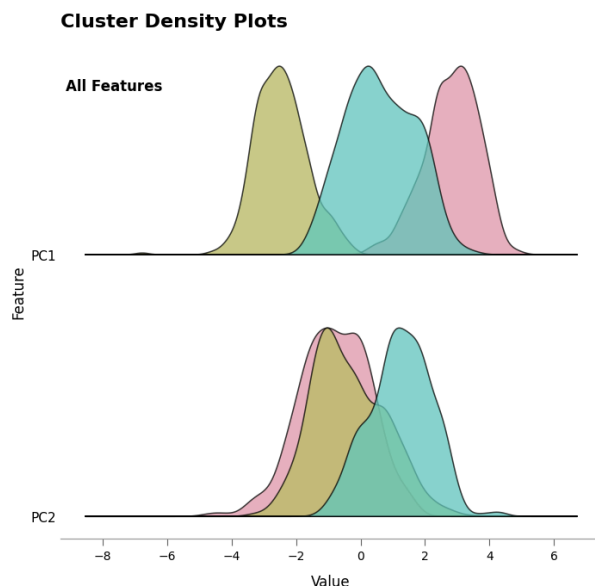
Although some boundary proximity is visible between neighboring groups, the overall structure suggests that the final K-means solution achieved **meaningful segmentation**.

Cluster Means		
	PC1	PC2
Cluster 1	2.808	-0.814
Cluster 2	-2.468	-0.374
Cluster 3	0.615	1.235



The cluster mean plots indicate clear differences among the three customer segments across the retained PCA dimensions. Cluster 1 demonstrated the highest scores on PC1, suggesting strong **customer purchasing power**, while Cluster 2 showed negative scores across both components, indicating comparatively lower purchasing activity and engagement. In contrast, Cluster 3 exhibited the highest PC2 scores, reflecting more pronounced **household-driven and deal-oriented purchasing behavior**.

These mean profiles support the interpretability of the final K-means clustering solution.



The cluster density plots support the distinctiveness of the three customer segments, as each cluster demonstrates **relatively concentrated distributions** across PC1 and PC2 with only moderate overlap between groups.

The separation is particularly evident along PC1, reinforcing the adequacy and interpretability of the final K-means clustering solution.

Customer Segmentation

Cluster 1 - Affluent High-Value Customers

Cluster 1 represents affluent and highly engaged customers characterized by strong purchasing power, high spending activity, and low dependence on promotional deals.

cluster	numwebvisitsmonth	totalspending	totalpurchases	familysize	spendingincomeratio
1	2.910000	1300.880000	19.670000	1.850000	0.020000

This cluster exhibited the highest total spending and total purchases despite relatively fewer monthly web visits. The group also showed the highest spending-to-income ratio, suggesting strong purchasing power and active customer engagement. Family size was relatively smaller compared to the other clusters.

cluster	webconversionrate	dealpurchaseratio	age	income	education
1	1.520000	0.070000	52.670000	74336.530000	2

Affluent high-value customers demonstrated the highest web conversion rate and the lowest deal purchase ratio, indicating efficient purchasing behavior with low dependence on promotions or discounts. This segment also had the highest average income and an education level corresponding to a bachelor’s degree, suggesting a financially capable and well-educated customer group.

cluster	marital_status	costumertenuredays	complain	campaignacceptancerate
1	2.050000	349.350000	0.000000	0.150000

These people demonstrated the highest campaign acceptance rate (0.150) and no recorded complaints, suggesting strong customer satisfaction and responsiveness to marketing efforts. Although customer tenure was slightly shorter than Cluster 3, the group still appeared highly engaged with the business.

Cluster 2 - Low-Value Deal-Driven Browsers

Cluster 2 consists of lower-spending and highly deal-sensitive customers who frequently browse online but demonstrate relatively weak purchasing conversion.

cluster	numwebvisitsmonth	totalspending	totalpurchases	familysize	spendingincomeratio
2	6.600000	77.340000	5.330000	2.900000	0.000000

This cluster of customers exhibited the highest number of monthly web visits but the lowest total spending and total purchases among the three segments. The group also showed a near-zero spending-to-income ratio, suggesting comparatively weak purchasing activity despite high browsing behavior. Family size was relatively larger compared to Cluster 1.

cluster	webconversionrate	dealpurchaseratio	age	income	education
2	0.250000	0.340000	48.360000	32435.400000	2

This segment demonstrated the lowest web conversion rate and the highest deal purchase ratio, indicating strong dependence on promotions and discounts. It also had the lowest average income, while the average education level corresponded to a bachelor's degree.

cluster	marital_status	costumertenuredays	complain	campaignacceptancerate
2	1.960000	326.880000	0.010000	0.030000

Compared to the others, Cluster 2 showed the lowest campaign acceptance rate (0.030) and the shortest customer tenure among the three segments, indicating comparatively weaker engagement with the business overall. Complaint incidence remained low despite the group's lower purchasing activity.

Cluster 3 - Stable Moderate-Value Family Customers

Cluster 3 reflects mature and household-oriented customers with moderate-to-high purchasing activity and relatively stable spending behavior.

cluster	numwebvisitsmonth	totalspending	totalpurchases	familysize	spendingincomeratio
3	5.880000	654.610000	15.400000	2.890000	0.010000

This set of customers demonstrated moderate-to-high total spending and purchase activity, with values lower than Cluster 1 but substantially higher than Cluster 2. The group also showed relatively high monthly web visits and larger family sizes, suggesting more active household-oriented purchasing behavior. The spending-to-income ratio remained moderate compared to the other clusters.

cluster	webconversionrate	dealpurchaseratio	age	income	education
3	0.850000	0.230000	56.410000	56622.590000	3

Stable family customers also exhibited moderate web conversion and deal purchase ratios, indicating balanced purchasing behavior with some sensitivity to promotions. This segment also had relatively high income levels, the oldest average age among the clusters, and an education level corresponding to postgraduate attainment, suggesting a mature and well-educated customer group.

cluster	marital_status	costumertenuredays	complain	campaignacceptancerate
3	1.890000	391.750000	0.010000	0.070000

They exhibited the longest customer tenure (391.75 days) and very low complaint incidence, suggesting a stable and established customer segment with generally positive customer experiences. However, the campaign acceptance rate remained moderate relative to Cluster 1.